LESSON PLAN

GUIDED WEAPON (37)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Terminal ballistics, type of damage, perforation effect, critical velocity
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Terminal ballistics, type of damage, perforation effect, critical velocity. Experimental ballistics – measurement of pressure and velocity
5. Specific Objectives	37.1.1.1 Trainee will be able to explain about various branches of ballistics

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of experimental ballistics data; Safety: Handle charts carefully; Supervision: Instructor monitors data interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four main branches of ballistics?
(b) Define terminal ballistics in one sentence.
(c) List two factors affecting projectile penetration.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of terminal ballistics • Types of damage caused by projectiles • Mechanism of perforation effect • Concept of critical velocity • Experimental measurement methods	L & Ds	PPT, IB	15 min
(b)	• Analysis of pressure and velocity data • Factors influencing perforation depth • Role of material strength in damage • Calculation of critical velocity thresholds • Interpretation of experimental results	L & Ds	PPT, PDF	20 min
(c)	• Summary of terminal ballistics concepts • Application of critical velocity in design • Review of experimental measurement techniques • Link between damage types and velocity • Preparation for next lesson on rocket ballistics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how critical velocity determines perforation success.
(b) Describe the relationship between projectile velocity and damage type.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of terminal ballistics discussed in this lesson apply to the stability forces covered in LP 05.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Rocket ballistics introduction, forces acting on rocket in flight and stability	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min